

Unit 4: Searching for Geographic Objects

4.1 Finding Information in Raster Systems

In raster systems, spatial data is stored in grid cells, each holding a value representing specific information (e.g., temperature, elevation, land cover). Finding information involves analyzing these cell values to extract meaningful insights.

Concepts of Finding Information in Raster Systems:

- **Cell-Based Queries:**
Users can identify information about a specific location by clicking on a cell to retrieve its attribute (e.g., land use type, elevation).
- **Map Algebra:**
A method to perform mathematical operations on raster layers to generate new data (e.g., calculating slope from elevation).
- **Zonal Statistics:**
Analyzes values within defined zones (e.g., average rainfall within a watershed).
- **Reclassification:**
Assigns new values to raster cells based on defined criteria (e.g., grouping elevation ranges into high, medium, low).
- **Overlay Analysis:**
Combines multiple raster layers (e.g., land cover + rainfall) to analyze relationships or suitability.

4.2 Finding Features in Vector Systems

In vector systems, spatial features are represented as points, lines, and polygons. Finding features involves querying these geometries and their attributes to retrieve spatial and non-spatial information.

Concepts of Finding Features in Vector Systems:

- **Attribute Query (SQL-based):**
Uses expressions like `SELECT * FROM roads WHERE type = 'highway'` to find features based on attribute values.
- **Spatial Query:**
Finds features based on spatial relationships (e.g., within, intersects, contains).
Example: Find all buildings within 100 meters of a river.
- **Identify Tool:**
In GIS tools like QGIS, clicking a feature shows detailed attribute data.
- **Select by Location:**
Selects features based on their spatial relationship with another layer (e.g., parks within a city boundary).

- **Layer Filtering:**

Temporarily displays only features that match a condition (e.g., show only schools in Kathmandu).

4.3 Searching Polygons in a GIS

Polygons in GIS represent area-based features like districts, lakes, forests, or land parcels. Searching polygons involves identifying, querying, and analyzing these features based on spatial or attribute criteria.

Techniques to Search Polygons:

1. **Attribute Query (SQL):**

Search polygons using descriptive data.

Example:

```
SELECT * FROM land_parcel WHERE zoning = 'residential'
```

2. **Spatial Query:**

Find polygons based on their spatial relationship with other features.

Examples:

- Find all polygons that intersect with a road.
- Select land parcels within a flood zone using ST_Within() in PostGIS.

3. **Select by Location:**

In GIS software, use spatial selection tools to find polygons within, intersecting, or containing other layers.

4. **Filter Expressions:**

Apply expressions to view only polygons that match certain conditions (e.g., area > 5000 sq. meters).

5. **Using GIS Tools (e.g., QGIS):**

- Use the **Identify Features** tool to click and retrieve polygon data.
- Use **Layer Styling Panel** to highlight or label specific polygons dynamically.

Applications:

- Finding administrative zones with a population over 10,000
- Selecting conservation areas that overlap with mining zones
- Identifying agricultural plots under government subsidy programs

4.4 Locating 2-D Map Objects

Locating 2-D map objects refers to identifying and pinpointing spatial features (points, lines, polygons) on a two-dimensional map within a GIS environment. These objects represent real-world entities like buildings, roads, or land parcels.

Techniques for Locating 2D Map Objects:

1. Coordinate-Based Location:

- Use latitude and longitude (or projected coordinates) to locate specific features on a map.
- Example: Entering coordinates (27.7172° N, 85.3240° E) to find a location in Kathmandu.

2. Attribute Search:

- Locate objects by querying attribute data.
- Example: Search for “Hospital” in a healthcare layer.

3. Spatial Indexing:

- GIS uses spatial indexes (e.g., R-tree) to quickly locate features in large datasets.

4. Identify Tool in GIS Software:

- Allows clicking on the map to retrieve information about any 2D object (e.g., in QGIS or ArcGIS).

5. Zoom and Pan Functions:

- Navigate the map to locate and inspect objects manually.

6. Labeling and Symbology:

- Use labels and distinct symbols to make 2D objects easier to identify visually.

Applications:

- Locating property parcels in cadastral mapping
- Finding roads, rivers, or public utilities in urban planning
- Navigating and analyzing infrastructure in emergency response systems

4.5 Defining the Groups for Searching

Defining groups for searching in GIS means organizing spatial features into logical sets or categories based on shared attributes or spatial characteristics to enhance query efficiency and analysis.

Purpose of Grouping in GIS Searches:

- Simplifies complex datasets
- Enhances filtering and selection processes
- Enables targeted analysis and visualization
- Supports thematic mapping and decision-making

Ways to Define Groups in GIS:

1. Attribute-Based Grouping:

- Features are grouped based on shared values in attribute fields.
- *Example:* Grouping land parcels by land use type (e.g., residential, commercial, agricultural).

2. Spatial-Based Grouping:

- Group features based on location or proximity.
- *Example:* All schools within 1 km of a highway.

3. Layer-Based Grouping:

- Group features within the same data layer or thematic map.
- *Example:* All rivers in a hydrography layer.

4. Symbolic Grouping (Symbology):

- Visual grouping using styles like color, shape, or size to represent categories.
- *Example:* Differentiating risk zones using color gradients.

5. User-Defined Sets or Queries:

- Use SQL-like expressions to define custom groups.
- *Example:* "Population" > 10000 AND "AreaType" = 'Urban'

Applications:

- Environmental monitoring (e.g., grouping areas with similar vegetation)
- Urban development (e.g., zoning categories)
- Emergency planning (e.g., high-risk zones)
- Resource management (e.g., water bodies grouped by volume)